

Experiment – 1

Title :

Library Borrowing Record Management using Python

Aim :

To develop a Python program for managing library borrowing records and perform statistical operations such as average, maximum, minimum, zero borrow count, and mode using Hungarian notation.

Problem Statement :

Write a Python program to manage the borrowing records of books in a library. Implement the following functionalities:

- Compute the average number of books borrowed by all library members.
- Find the highest and lowest number of borrowings.
- Count the number of members who have not borrowed any books (borrow count = 0).
- Display the most frequently borrowed count (Mode).
- Determine the Time and Space Complexity of each operation.

Algorithm :

Step 1 - Start.

Step 2 - Read the number of library members.

Step 3- Create an empty list.

Step 4- Accept borrow count of each member.

Step 5- Store each borrow count in the list.

Step 6- Calculate the average borrow count.

Step 7 - Find the highest borrow count.

Step 8- Find the lowest borrow count.

Step 9 - Count members whose borrow count is zero.

Step 10- Find the frequency of each borrow count.

Step 11- Display the borrow count having maximum frequency (Mode).

Step 12 -Display all results.

Step 13 -Stop.

Code:

```
# Library Borrowing Records
```

```
n = int(input("Enter the number of library members: "))
```

```
# Create a list of size n
```

```
borrow = [0] * n
```

```
# Input borrow counts
```

```
for i in range(n):
```

```
    borrow[i] = int(input("Enter books borrowed by member " + str(i + 1) + ": "))
```

1. Calculate Average

total = 0

for i in range(n):

total = total + borrow[i]

average = total / n

print("\nAverage books borrowed:", average)

2. Find Highest and Lowest Borrow Count

highest = borrow[0]

lowest = borrow[0]

for i in range(1, n):

if borrow[i] > highest:

highest = borrow[i]

if borrow[i] < lowest:

lowest = borrow[i]

print("Highest borrow count:", highest)

print("Lowest borrow count:", lowest)

3. Count Members with Zero Borrowings

zero_count = 0

for i in range(n):

if borrow[i] == 0:

```
zero_count = zero_count + 1
```

```
print("Members who borrowed no books:", zero_count)
```

```
# 4. Find Mode (Most Frequent Borrow Count)
```

```
mode = borrow[0]
```

```
max_count = 0
```

```
for i in range(n):
```

```
    count = 0
```

```
    for j in range(n):
```

```
        if borrow[i] == borrow[j]:
```

```
            count = count + 1
```

```
    if count > max_count:
```

```
        max_count = count
```

```
        mode = borrow[i]
```

```
print("Most frequently borrowed count (Mode):", mode)
```

Output :

```
Enter the number of library members: 6
Enter books borrowed by member 1: 10
Enter books borrowed by member 2: 5
Enter books borrowed by member 3: 3
Enter books borrowed by member 4: 10
Enter books borrowed by member 5: 6
Enter books borrowed by member 6: 4

Average books borrowed: 6.333333333333333
Highest borrow count: 10
Lowest borrow count: 3
Members who borrowed no books: 0
Most frequently borrowed count (Mode): 10
```

